Roll No.

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science
Auxiliary Exam- Feb, 2026

B.E. (4th Year) **Course Code:** UCS802 **Course Name:** Compiler Construction
Instructor Name: Dr. Shashank Singh Max Marks: 100

Instructions: Attempt any **four** questions. Assume missing data suitably, if any.

Q1.	**a)** Discuss the different phases of compilation. Explain the answer with suitable example.
	
	b) Find number of tokens in following code snippet
	int max (i, j)
	int i, j ;
	/* return max of i && j */
	{
	return i > j ? i : j;
	}
	c) Consider the following code fragment and identify the lexical errors
	int main() {
	int 9count = 10;
	float value = 12.34.56;
	char ch = 'ab';
	string s = "Hello World;
	@total = value + #count;
	if (value > 100) {
	printf("Valid\n");
	}
	}
		10+ 7+8

Q2.	**a)** Discuss left recursion and left factoring. Also explain why it is essential to remove left recursion and left factoring in top-down predictive parsing. Write the rules to remove the left recursion and left factoring. **b)** Check if the following grammar is LL(1) or not S-> aBDh B-> cC C-> bC/ε D -> EF E -> g/ ε F -> f/ ε	10 +15
Q3.	**a)** What do you mean by bottom-up parsing. Discuss different types of bottom-up algorithms. **b)** Determine whether the following grammar is LR(0). If it is not LR(0), identify the number of inadequate states in its LR(0) parsing table. $S \rightarrow SS \mid a \mid \varepsilon$	10+15

Q4.	**a)** Describe and differentiate between synthesized attribute and inherited attributes. Explain your answer with suitable example. **b)** Prepare a SDT to convert the given infix to postfix expression i/p = a+b*c o/p = +a*bc **c)** Construct syntax directed translation to convert binary number in to decimal number Input = 101.101 Output = 5.625	7+8+10

Q5	**a)** What are the different types of acceptable 3-address codes. Convert the following code snippet into 3-address code format using back patching technique.
	
	fact(x)
	{
	int f =1;
	for (i=2; i < x; i++)
	f = f*i;
	return f;
	}
	
	b) What are the different intermediate codes. Discuss in detail.
		13+12